Example Co.

Gage R&R Study — ANOVA

27 Oct 2026

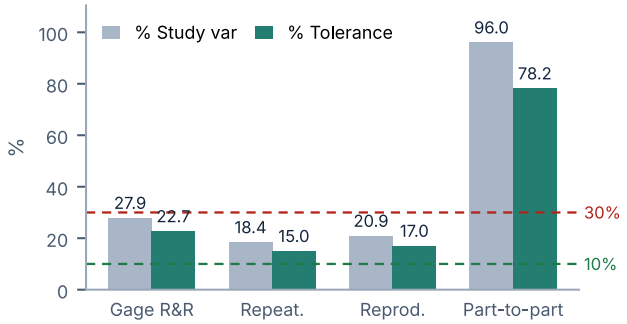
PART NUMBER **SAMPLE-001**	CHARACTERISTIC **AIAG MSA 4th ed. worked example**	SPECIFICATION **LSL -4 · USL 4**
GAGE **Example gage**	STUDY DESIGN **10 parts · 3 operators · 3 trials**	CUSTOMER Not given

CONDITIONALLY ACCEPTABLE

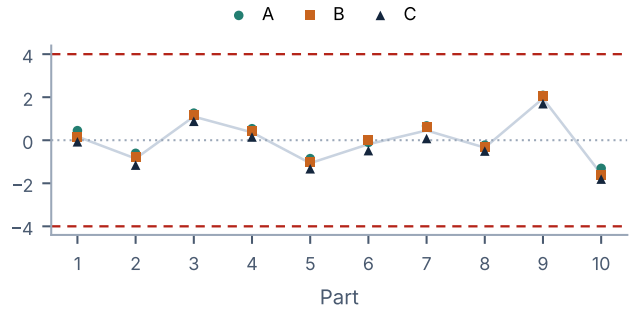
%GRR is **22.7% of tolerance**, inside the 10-30% band. Use may be accepted depending on the application, cost and customer approval. The gage resolves only 4 distinct categories of parts; at least 5 are needed. Operator B shows the widest repeat-reading ranges (average range 0.513).

%GRR · TOLERANCE	%GRR · STUDY VAR	DISTINCT CATEGORIES	PART × OPERATOR
22.7%	**27.9%**	**4**	**p = 0.974**
Limits 10% / 30%	Of total variation	Required ≥ 5 · fail	Not significant · pooled

Components of variation



Average by part and operator



Source	Std dev	% Study var	% Tolerance	% Contribution
Total Gage R&R	**0.302**	**27.86**	**22.68**	**7.76**
Repeatability (EV)	0.200	18.42	14.99	3.39
Reproducibility (AV)	0.227	20.90	17.01	4.37
Part-to-part (PV)	1.042	96.04	78.17	92.24
Total variation (TV)	**1.085**	**100.00**	**81.40**	**100.00**

Prepared by · Sample report

Approved by

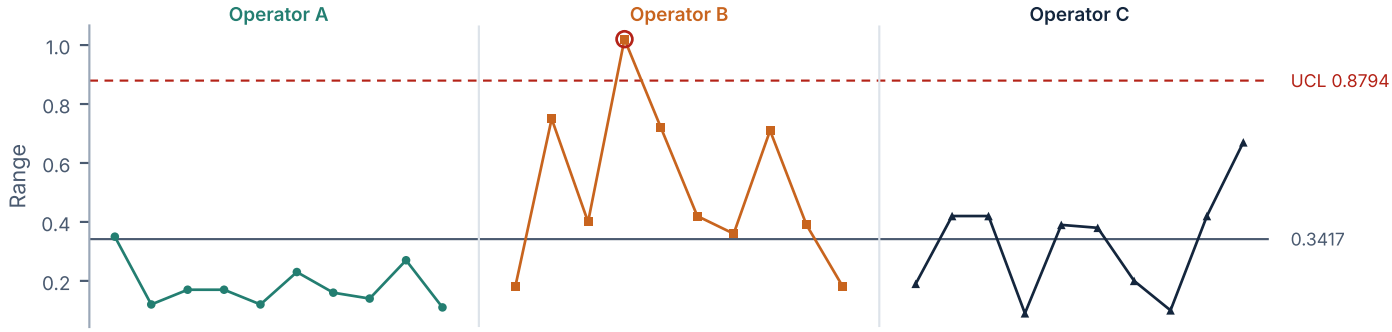
Date

Charts by operator

10 parts · 3 operators · 3 trials

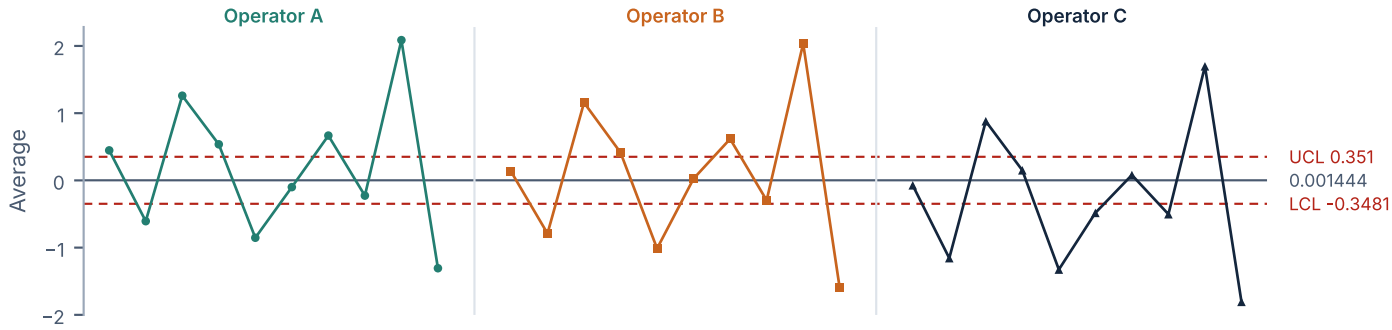
Range chart

Each point is the range of one operator's repeat readings on one part. Limits: $\bar{R}$ = 0.3417, UCL = $D_4\bar{R}$ = 0.8794 (D_4 = 2.574 for 3 trials).



Average chart

Each point is one operator's average for one part. Limits: $\bar{\bar{X}}$ = 0.0014444 ± $A_2\bar{R}$ (A_2 = 1.023), so UCL = 0.35097 and LCL = -0.34808.



What the charts show

- 1 range(s) are above the range chart limit (operator B on part 4). Check those readings for a measuring or recording problem before relying on the study.
- 73% of the part averages fall outside the average chart limits. That is expected: the gage can tell the parts apart (AIAG MSA looks for about half or more).

ANOVA table

Source	DF	SS	MS	F	p
Part	9	88.362	9.8180	245.61	< 0.001
Operator	2	3.1673	1.5836	39.62	< 0.001
Repeatability	78	3.1179	0.039973		
Total	**89**	**94.647**			

The part × operator interaction is not significant (p = 0.974 > α = 0.05), so it is pooled into repeatability.

Variance components

Source	Variance	% Contribution
Repeatability (EV)	0.040	3.39
Reproducibility (AV)	0.051	4.37
Total Gage R&R	**0.091**	**7.76**
Part-to-part (PV)	1.086	92.24
Total variation (TV)	**1.178**	**100.00**

Method and decision rules

How every number in this report was made

Method

Two-way crossed ANOVA with parts and operators as random factors, following the AIAG Measurement Systems Analysis Reference Manual, 4th edition. The part × operator interaction is tested at α = 0.05 and pooled into repeatability because it is not significant.

$\sigma^2_{EV} = MS_{repeatability}$ (error and interaction pooled)
$\sigma^2_{AV} = (MS_{operator} - MS_{repeatability}) / (n\ r)$
$\sigma^2_{PV} = (MS_{part} - MS_{repeatability}) / (k\ r)$

n = 10 parts, k = 3 operators, r = 3 trials. A negative variance estimate is set to zero.

$GRR^2 = EV^2 + AV^2 \quad TV^2 = GRR^2 + PV^2$
% Study var = $100 \times \sigma / \sigma_{TV}$ % Tolerance = $100 \times 6\sigma$ / tolerance
% Contribution = $100 \times \sigma^2 / \sigma^2_{TV}$
ndc = $1.41 \times PV / GRR$, rounded down

Charts by operator

Range chart limits $D_3\bar{R}$ and $D_4\bar{R}$; average chart limits $\bar{\bar{X}} \pm A_2\bar{R}$, using $\bar{R}$ and $\bar{\bar{X}}$ across all operators and constants for subgroups of 3 (A_2 = 1.023, D_3 = 0.0, D_4 = 2.574; Montgomery, Introduction to Statistical Quality Control, 6th ed., Appendix VI).

Decision rules

%GRR is judged against the tolerance (AIAG MSA 4th ed.).

- **Under 10%**: acceptable.
- **10% to 30%**: may be acceptable depending on the application, cost and customer approval.
- **Over 30%**: not acceptable.

The number of distinct categories (ndc) should be 5 or more; below 5, the study is at best conditionally acceptable. Your customer's requirements apply if they differ.

Receipt

Method	two-way crossed ANOVA (AIAG MSA 4th ed.)
Study design	10 parts × 3 operators × 3 trials = 90 readings
ndc factor	1.41
Study variation (σ multiple)	6.0
Tolerance	8.0
Interaction α	0.05
Verdict basis	tolerance
Engine	Mensuro engine 0.1.0
Generated	2026-10-27

- Part x operator interaction p = 0.974; not significant, so it was pooled into repeatability.

How the math is checked

Sums of squares come from statsmodels and F probabilities from SciPy. Before release, Mensuro's Gage R&R engine must reproduce published reference results: the AIAG MSA 4th edition worked example and Montgomery's Table 8.7 example (with a significant interaction). The range and average chart limits reproduce Montgomery Example 8.7. These checks rerun on every change.

Data

All 90 readings

Part	Operator A					Operator B					Operator C				
	Trial 1	Trial 2	Trial 3	Average	Range	Trial 1	Trial 2	Trial 3	Average	Range	Trial 1	Trial 2	Trial 3	Average	Range
1	0.29	0.41	0.64	0.447	0.35	0.08	0.25	0.07	0.133	0.18	0.04	-0.11	-0.15	-0.073	0.19
2	-0.56	-0.68	-0.58	-0.607	0.12	-0.47	-1.22	-0.68	-0.790	0.75	-1.38	-1.13	-0.96	-1.157	0.42
3	1.34	1.17	1.27	1.260	0.17	1.19	0.94	1.34	1.157	0.40	0.88	1.09	0.67	0.880	0.42
4	0.47	0.50	0.64	0.537	0.17	0.01	1.03	0.20	0.413	1.02	0.14	0.20	0.11	0.150	0.09
5	-0.80	-0.92	-0.84	-0.853	0.12	-0.56	-1.20	-1.28	-1.013	0.72	-1.46	-1.07	-1.45	-1.327	0.39
6	0.02	-0.11	-0.21	-0.100	0.23	-0.20	0.22	0.06	0.027	0.42	-0.29	-0.67	-0.49	-0.483	0.38
7	0.59	0.75	0.66	0.667	0.16	0.47	0.55	0.83	0.617	0.36	0.02	0.01	0.21	0.080	0.20
8	-0.31	-0.20	-0.17	-0.227	0.14	-0.63	0.08	-0.34	-0.297	0.71	-0.46	-0.56	-0.49	-0.503	0.10
9	2.26	1.99	2.01	2.087	0.27	1.80	2.12	2.19	2.037	0.39	1.77	1.45	1.87	1.697	0.42
10	-1.36	-1.25	-1.31	-1.307	0.11	-1.68	-1.62	-1.50	-1.600	0.18	-1.49	-1.77	-2.16	-1.807	0.67

Operator	Average ($\bar{X}$)	Average range ($\bar{R}$)
A	0.1903	**0.1840**
B	0.0683	**0.5130**
C	-0.2543	**0.3280**